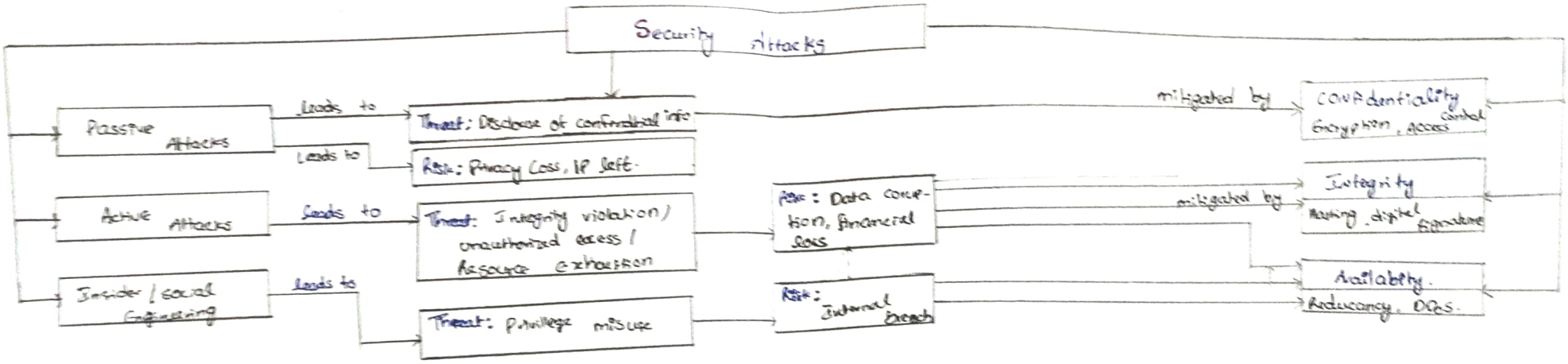
1)

Security Attacks → Threats → Risks ↔ Security Goals

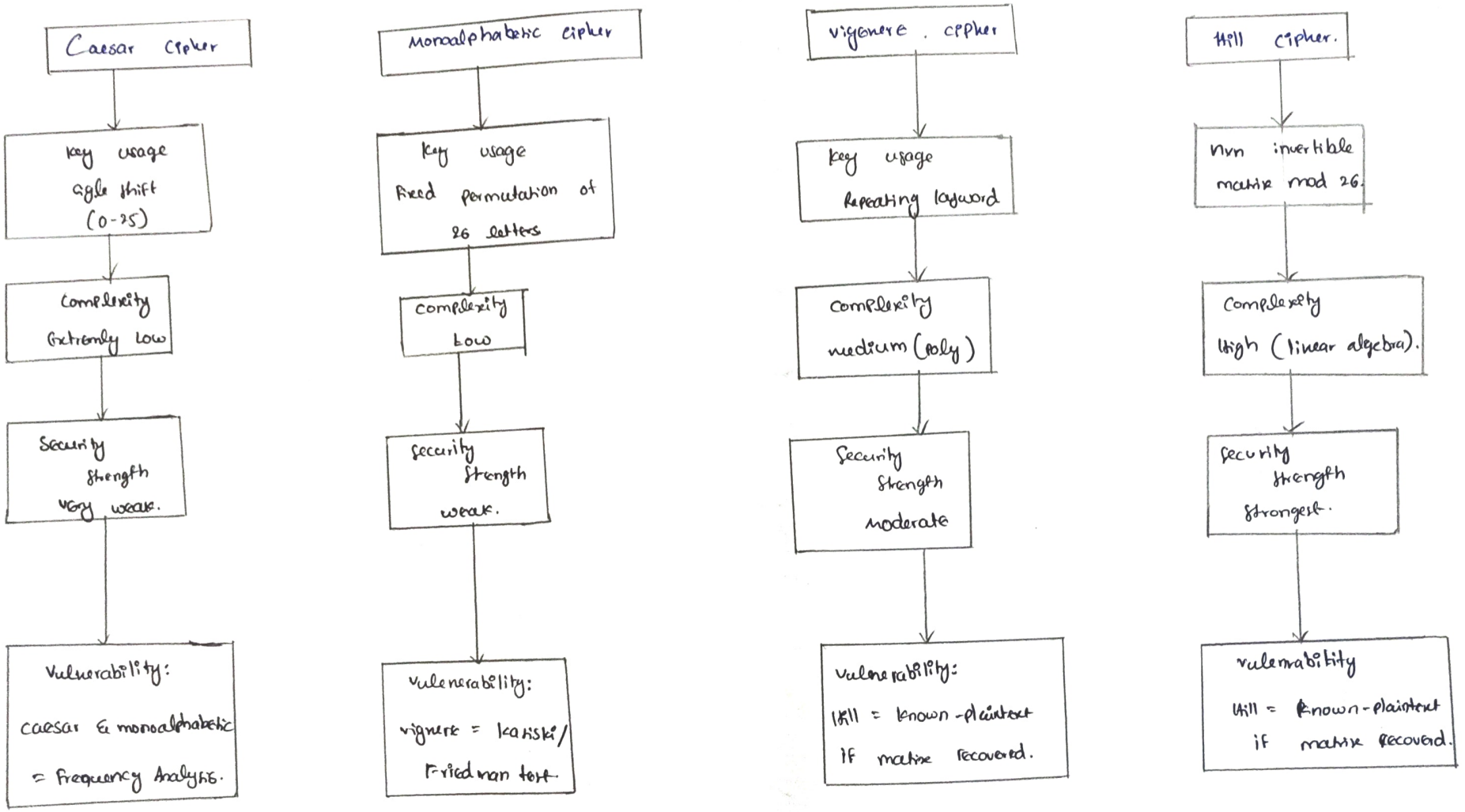
(CIA)



Attack	Threat	Risk	Goal	Real-world Example:
Eavesdropping	Data confidentiality breach	Privacy violation	Confidentiality	Target 2013 POS breach
Message modification	Integrity violation	Fraud	Integrity	Bangladesh Bank SWIFT 2016.
DoS	Availability Loss	Downtime	Availability	Mirai Dyn DNS 2016.
Masquerade	Unauthorized access	Account takeover	Integrity + Auth	Token replay Attacks.
Traffic Analysis	Pattern disclosure	Business interference.	Confidentiality.	VPN traffic monitoring.

2.

Substitution Techniques Comparition: Caesar | monoalphabetic | vigenere | Hill. cipher.



8.

Classical Encryption Schemes: Substitution Vs Transposition.

Substitution Techniques	Differences / Similarities	Transposition Techniques.
• Caesar cipher • monoalphabetic cipher. • Vigenere cipher • Hill cipher • playfair	Differences.	• Rail fence. cipher. • Columnar cipher • Route • Double Transpotion.
• Replace symbols with other symbols. • Data structure: content changes, order same. • Distorts letter frequencies. • Vulnerable to frequency analysis.	Similarities Both are classical cipher building blocks, both permute or transform plaintext.	• Rearrage order of symbol. • Data structure: order changes, content same. • Preserve letter frequencies. • Vulnearable to anagramming & pattern Attacks.

Aspect	Substitution	Transportion.
Operation	Replace symbols with other symbols	Rearrange order of symbols.
Effect on data	content changes, order same.	order changes, content same.
Frequency statistics.	Disorts letter freauencies.	Preserves letter frequencies.
Typical attacks.	Frequency analysis & Kasiki	Anagramming & pattern Attacks.
combining both	Forms product ciphers	Forms product ciphers (much stronger).

4.

Steganography Concepts : Data Hiding. water marking. Survivability.

Data Hiding.

Goal:- Conceal existence of secret message.

Techniques:- LSB. DCT phase embedding.

Contribution: covert communication & deniability.

water marking

- Goal: Embed ownership

Types: Robust vs Fragile. visible vs invisible.

Contribution:

Relationships.

Survivability.

Ability of hidden data to resist attacks (compression cropping, noise, formal covert) Links the other two: without Survivability both become useless.

concept	contribution to secure Communication	Real world Application.
Data Hiding.	Enable covert channels.	Secret message in social media images.
watermarking	Provides ownership proof.	Digmarc in images / music / broadcast monitoring.
Survivability.	Ensures Survival though Processing.	Robust video watermarks surviving youtube compressing, medical DICOM stegenography.